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School of [School Name]

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University of Lincoln

Submitted in partial fulfilment of the requirements for the Degree of [Your Programme Title e.g. Master of Science in Computing]

Primary Supervisor: [Primary Supervisor Name]
Secondary Supervisor: [Secondary Supervisor Name]

[Month Year]

Acknowledgements

Write your acknowledgements here. You may wish to thank your supervisors, participants, friends, and family.

Abstract

Write your abstract here. Summarise the research problem, methodology, key findings, and conclusions.

Keywords: [Keyword 1], [Keyword 2], [Keyword 3], [Keyword 4], [Keyword 5]

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Glossary

Add glossary terms here, arranged alphabetically. Define key terms used in your dissertation to ensure clarity and consistency.

Example Term – Definition of the term as it is used in your dissertation.

Chapter 1

Introduction

The introduction should give an excellent overview of the project background and set the scene for the reader. It should establish the context and significance of your work, and provide a clear motivation for why this project is worth undertaking. By the end of this chapter, the reader should understand the problem space and be eager to learn more.

This document is a project report template for the School of Computer Science at the University of Lincoln. It should give you some direction and instruction for formatting and presenting your project report. This template is designed for postgraduate project reports, however, templates are also available for undergraduate dissertations and doctoral theses on the same GitHub repository.

This document has been generated using Typst, a modern typesetting system. The following subsections of this chapter demonstrate some basic Typst commands and formatting features.

1.1 Background and Motivation

Introduce the broad topic area and explain why it is important or relevant. This section should provide enough context for a reader unfamiliar with the domain to understand the significance of your work. You may wish to include a brief historical overview of key developments in the field to show how the current state of the art has been reached.

1.2 Subheading

1.3 Sub-subheading

1.4 Formatting Mathematics

Here are two equations:

$$a = b + 1$$

$$\frac{\hbar^2}{2m}\nabla^2\Psi + V(\boldsymbol{r})\Psi = -i\hbar\frac{\partial\Psi}{\partial t}$$

And here is some text with some nice inline maths, (x, y) wow γ so cool ρ.

1.5 Figures and Tables

Here is a sentence, and you can see a nice picture in Fig. 1.

Fig.1. A picture of the Brayford from Google Images.

Also, a table can be found in Table 1.

Table1. Here is a table. The caption goes above like this.

First Name	Last Name	Age
Bob	Bobbington	24
Benth	Wavies	49
Joe	Bloggs	37
Billy	Bob	10

1.6 Referencing

The included Harvard citation style file means that referencing will follow a Harvard style. You can cite references inline using the `@citekey` syntax. For example, here is a citation (Smith and Doe, 2024). You can also reference conference papers (Williams and Johnson, 2023) and books (Brown, 2022).

It is worth noting that the standard for referencing is Harvard, so you will not need to change this option. However, please double and triple check this to make sure you are using the correct referencing style. Also, ask your supervisor to make sure. This template uses a BibTeX (`.bib`) file for referencing. It is recommended to use a reference manager such as Zotero to export your references as a `.bib` file.

Chapter 2

Literature Review

The literature review is an essential requirement of any academic project. A comprehensive review of the literature will provide background to the project, and should be used to inform a set of requirements that your solution must meet. It also establishes that what you have done is the result of academic study, rather than an unfounded whim. This should be a coherent, critical evaluation of relevant academic literature with clear linkage to your project.

It may be helpful to break up this chapter into sections, with each focused on a different topic or aspect of the project. At MSc level, you are expected to provide a comprehensive and insightful review, with an impressive degree of critical evaluation that clearly informs the final outcomes of the project.

2.1 The Research Gap

Having reviewed the literature, you should identify a clear gap in knowledge or practice that your project addresses. This section should synthesise the evidence from your review and articulate the specific problem or opportunity that motivates your work.

2.2 Aim and Objectives

Having situated your project within a body of relevant literature, you should now be in a position to state your aims and objectives. Most projects will have one aim that is a broad statement of what the project will achieve. The objectives should be statements of how that aim will be achieved. Objectives should be Specific, Measurable, Achievable, Realistic and Time-related (SMART).

2.2.1 Aim

State the overarching aim of your project in a single, clear sentence.

2.2.2 Objectives

List your SMART objectives here. These should be concrete, measurable steps that collectively achieve the aim.

2.2.3 Research Questions and Hypotheses

If your project involves a user study or empirical research, you should state your research questions here. If appropriate, you may also state formal hypotheses. Research questions should flow naturally from the research gap identified above.

Chapter 3

Requirements Analysis

Drawing upon the research you have conducted, the goal of this chapter is to formalise a set of requirements that your project must fulfil in order to meet your objectives. This should expand upon the aims and objectives, stating what your solution must do to achieve the desired project outcomes. This chapter can be relatively short, and reference the aim(s) and objectives given at the end of the previous chapter.

Consider both functional and non-functional requirements, software/hardware environments, a risk analysis, and briefly how you will test and evaluate your artefact. For software-oriented projects, consider key stakeholders (end-users, business owners) when analysing the requirements of your software. This may be through primary research (i.e., interviews with potential end-users or clients) or secondary research (e.g., literature review, survey of existing products). For research-oriented projects, use your literature survey to highlight current gaps in knowledge that you hope to address. Consider what your artefact must do in order to obtain meaningful results.

Chapter 4

Design and Methodology

This section of the dissertation may vary significantly in both structure and content, depending on the type of project you are undertaking. Many MSc projects involve both a software artefact and a research component — in such cases, it is helpful to split this chapter into two distinct parts, as shown below. The precise structure should be discussed with your supervisor.

The key thing to note here is that irrespective of the project type, you should justify the choices you have made with academic reasoning, rather than simply choosing based on expediency or familiarity.

4.1 Part A: Research Methodology

If your project involves a user study or empirical research, this section should detail your research design. The following subsections are recommended:

4.1.1 Experimental Design

Describe and justify your research design (e.g., within-subjects, between-subjects, mixed methods). Explain your design rationale and why this approach is appropriate for answering your research questions.

4.1.2 Participants

Detail your approach to participant recruitment, including sample size determination (e.g., power analysis), recruitment strategy, and eligibility criteria (both inclusion and exclusion). If applicable, describe demographics collection methods.

4.1.3 Measures and Instruments

Describe the instruments and measures used to collect data (e.g., validated questionnaires, custom surveys, physiological sensors).

Justify the selection of each instrument with reference to the literature.

4.1.4 Procedure

Describe the step-by-step procedure that participants follow during your study, in enough detail that it could be replicated.

4.1.5 Data Analysis Plan

Describe how you plan to analyse the collected data. For quantitative data, specify the statistical tests you will use and why. For qualitative data, describe your analysis approach (e.g., thematic analysis).

4.1.6 Ethical Considerations

Describe the ethical considerations of your study and confirm that appropriate ethical approval has been obtained. Reference any ethics documentation in the appendices.

4.2 Part B: Software Development

This section covers the design and development methodology for the software artefact.

4.2.1 Software Engineering Methodology

Describe and justify the software development methodology used (e.g., agile, waterfall, scrum). Explain why this approach was suitable for the project.

4.2.2 Risk Analysis

Document the risks associated with the project and how they were mitigated. Consider both technical risks (e.g., hardware failure, software bugs) and project risks (e.g., time constraints, scope creep).

4.2.3 Toolsets and Environments

Detail the software tools, frameworks, libraries, and hardware environments used in the project, and justify their selection.

4.2.4 Design

Present the design of your software system (e.g., UML diagrams, database schemas, system architecture diagrams, wireframes, prototypes). For games projects, this may include a game design document.

4.2.5 Testing

Describe the types of testing used (e.g., unit testing, integration testing, user acceptance testing) and your approach to quality assurance.

Chapter 5

Implementation

This section should describe all components of your software implementation, emphasising particularly novel or nuanced components of code or its structure. There is no need to document every line of code, but you may include snippets as appropriate. You may wish to include code snippets and screenshots as figures to provide detail where necessary. At MSc level, an artefact with very high technical complexity is expected, and the implementation should be very well documented.

5.1 System Architecture and Technical Overview

Provide a high-level overview of the system architecture. This may include architecture diagrams, descriptions of major components and their interactions, and the technologies used. This gives the reader the big picture before diving into specifics.

5.2 Core Design and Mechanics

Describe the core design decisions behind your artefact. For software projects, this might cover key design patterns or architectural decisions. For games projects, this could include the core concept, gameplay loops, level design, key mechanics, and design goals.

5.3 Implementation Details

Document the most significant or technically interesting aspects of your implementation. This might be broken into subsections covering different subsystems or modules. Focus on what is novel, complex, or noteworthy — avoid documenting trivial or boilerplate code.

5.4 Testing and Quality Assurance

Describe the testing conducted during development (e.g., unit tests, integration tests, playtesting). Document any bugs found and how they were resolved.

5.5 Deployment and Performance

If applicable, describe how the software was deployed and any performance considerations or optimisations made.

5.6 Source Code and Documentation

Describe how the source code is organised and where it can be found. Note any documentation provided (e.g., README files, API documentation, user guides).

Chapter 6

Results and Discussion

This chapter should present the findings of your work, and discuss them in the context of your original aims and objectives. Present the data in an appropriate format (tables, charts, visualisations) and discuss what the data shows. You should link these results to your original objectives and discuss the extent to which they have been met, with clear evidence. At MSc level, results should be very well presented and clearly demonstrate the success of the project, with all original objectives met and supported by clear evidence.

6.1 Data Analysis Approach

Briefly describe how the data was processed and analysed. This links back to the data analysis plan in the methodology chapter, and confirms the approach taken.

6.2 Presentation of Results

Present your results clearly and systematically. Consider organising results by research question, hypothesis, or instrument, depending on what is most logical for your project.

6.2.1 Participant Demographics

If your project involved a user study, summarise the demographics of your participants (e.g., age, gender, experience level). This is typically presented as a table.

6.2.2 Quantitative Findings

Present any quantitative results, including statistical analyses. Use tables and charts to support the presentation. For each measure or instrument used, present the results and discuss their significance.

6.2.3 Qualitative Findings

If your project involved qualitative data collection (e.g., interviews, open-ended survey questions), present the results of your qualitative analysis here (e.g., thematic analysis, key themes and codes identified).

Chapter 7

Conclusion

The conclusion should provide a very detailed and insightful account of the achievements and limitations of the project, with excellent reflection on the development process and potential future directions.

7.1 Summary of Findings and Research Questions

Summarise the key findings of your research and explicitly address each of your original research questions. Where applicable, state whether your hypotheses were supported or rejected, and discuss the implications of these findings.

7.2 Achievement of Objectives

Reflect on whether your aims and objectives were met, and to what extent. Revisit each objective stated in the literature review and provide evidence of its completion or explain why it was not fully achieved.

7.3 Limitations

Discuss the limitations of your project, including any constraints that affected the scope, methodology, or results. Being honest about limitations demonstrates critical awareness and strengthens the academic quality of your work.

7.4 Future Work

Discuss potential directions for future research and any improvements that could be made. These should flow naturally from the

limitations identified above, as well as from any new questions raised by your findings.

References

Brown, R. (2022) *An Example Textbook*. Lincoln, UK: University Press

Smith, J. and Doe, J. (2024) An Example Journal Article. *Journal of Example Research,* 42(3) 100–115.

Williams, B. and Johnson, S. (2023) An Example Conference Paper. In: *Proceedings of the International Conference on Examples.* 2023 50–62

Appendices

This section is for any supplementary content that does not fit easily within the main dissertation. Examples include ethics approval, questionnaires, software documentation, or supplementary materials.

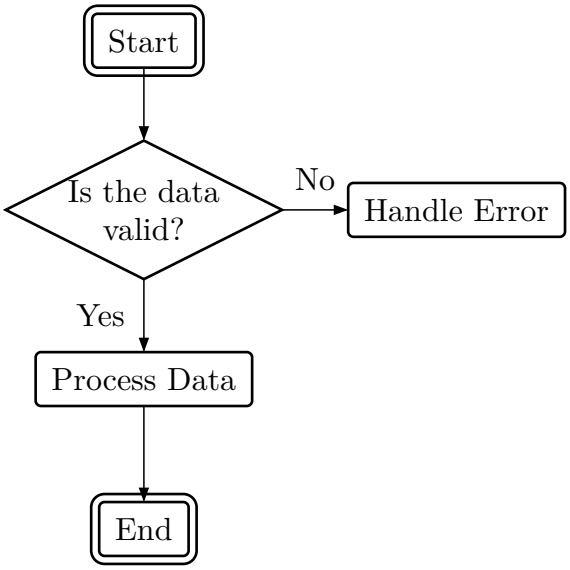
Appendix A: Code Snippets with Codelst

Here you can see a short snippet of code that was used to implement a simple Python sorting algorithm. The Codelst package provides syntax highlighting for code blocks:

```python
def bubble_sort(array):
    n = len(array)

    for i in range(n):

        already_sorted = True

        for j in range(n - i - 1):
            if array[j] > array[j + 1]:

                array[j], array[j + 1] = array[j + 1], array[j]

                already_sorted = False

        if already_sorted:
            break
    return array
```

Appendix B: Flowcharts with Fletcher

This uses the Fletcher package to create a flowchart. The package is available at: https://typst.app/universe/package/fletcher/



Word Count Summary

(excluding figures, tables, captions, and references)

Chapter	**Word Count**
Introduction	358
Literature Review	304
Requirements Analysis	164
Design and Methodology	427
Implementation	266
Results and Discussion	232
Conclusion	174
Total	**1925**

Overall Word Count: 1925 words